

A Novel Hybrid Plasmonic Waveguide for Nano-Scale Light Confinement and Long Propagation Range

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Abstract—A triangular-shaped metal insulator metal hybrid plasmonic waveguide has been proposed and optical properties of the fundamental hybrid mode at the wavelength of 1550 nm, such as propagation length, normalized effective mode area, etc., have been investigated in order to achieve the higher propagation length and to confine light at deep-subwavelength. The waveguide structure is mainly consisted of Silver (Ag) as a metal, Aluminum Gallium Arsenide (AlGaAs) as a high index region, and Polytetrafluoroethylene (PTFE) as a low index region. The electromagnetic wave is mainly confined at the top of the high-index region (AlGaAs) i.e., inside the low-index region (PTFE), enabling better performance compared to traditional waveguides. The performance analysis of this waveguide has been done by Finite Element Method (FEM) by adjusting the different parameters such as the waveguide's width, angle, and heights of the metal, high and low-index region. From the analysis of our research, results have been established that the propagation length of a fundamental hybrid mode of the presented waveguide $L_p \sim 359.01\mu\text{m}$ with the normalized effective mode area ~ 0.0246 at $w = 200\text{ nm}$, $h_{\text{AlGaAs}} = 300\text{ nm}$, $h_g = 10\text{ nm}$ and $h_m = 100\text{ nm}$. This waveguide is promising for nano-photonic applications, including power splitters, directional couplers, and optical biosensing devices.

Keywords— Surface plasmon polaritons. Photonic integration circuits. Hybrid plasmonic waveguide. Propagation length, Normalized effective mode area.

I. INTRODUCTION

Over the past ten years, integrated plasmonic devices have drawn much attention because of their tremendous potential for on-chip optical communication and data processing.

Surface plasmon polaritons (SPP), which are the interaction of the electromagnetic field and surface charges at a metal-dielectric interface, can significantly restrict light at the subwavelength scale and strengthen the local electric field. By confining light into subwavelength dimensions, nano-photonic devices have demonstrated enhanced performance, including lower power consumption, faster data transfer rates, and greater integration density. SPP-based waveguides offer promising solutions for highly compact, high-performance optical devices, making them ideal candidates for applications in communication, sensing, and integrated photonic systems [1], [2], [3]. Although the optical power loss in dielectric waveguides (DW) is essentially nonexistent, it is limited by diffraction. Due to the diffraction limit, the electromagnetic wave cannot propagate in this structure. Realizing the miniature optical waveguide and devices are thus a significant problem. Surface plasmon polaritons supported by plasmonic waveguides (PWs), can also be used to alleviate the issue of the diffraction limit [4], [5], [6]. A symmetric dielectric-loaded SPP waveguide is suggested in [11] for improved modal confinement and lower propagation loss. HPWs typically combine dielectric and metallic components, leveraging their complementary optical properties to achieve superior performance compared to conventional dielectric or metallic waveguides. Researchers have explored different designs, such as slot, wedge, and triangular-shaped HPWs, to optimize the trade-off between this structure's confinement capacity and propagation distance [7], [8], [9].

Recent studies have reported advancements in triangular-shaped HPWs, showcasing their potential for deep

subwavelength light confinement. However, silicon-on-insulator (SOI) based HPWs and optical devices are suffering from monolithic integration with active optical devices at optical communication wavelengths.

Therefore, we proposed AlGaAs based HPWs which exhibit a strong electro-optic effect, beneficial for fast optical signal routing, and have a large thermo-optic coefficient (two times larger than that of Si). For example, a triangular HPW with a combination of metal and high-index materials achieved a propagation length of 279.7 μm and a normalized effective mode area of 0.048 [10]. Moreover, the materials such as Polytetrafluoroethylene (PTFE), Aluminum Gallium Arsenide (AlGaAs), etc. are promising materials, capable of monolithic integration with active optical device [11], [12], [13][19], [23]. The material ‘AlGaAs’ has large nonlinear coefficients and negligible nonlinear absorption at 1.55 μm [24]. Furthermore, Coupled with the fact that it can produce the strongest nonlinear optical interaction for wavelength conversion, it exhibits low linear and nonlinear propagation losses in the Telecom spectral range (1400–1600 nm) which is highly desirable in a commercial context.

In this structure, Aluminum Gallium Arsenide (AlGaAs) is used as a high index region, Polytetrafluoroethylene (PTFE) is used as a low index region and Silver(Ag) is used as a metal. This simulation based analysis is done by the help of COMSOL Multiphysics on Finite Element method(FEM). At the time of designing HPWG we used different types of materials with proper metal height and width, the gap between metal and high index region was also considered for fabrication purposes. Also, provide different shapes in high index regions to get better propagation distance at the lower modal area.

In this waveguide, achieves a propagation length of approximately 359.01 μm and a normalized effective mode area of around 0.0246, representing significant improvements over traditional waveguides. For instance, the propagation length in this work (359.01 μm) surpasses the 279.7 μm reported in previous studies, while the normalized effective mode area (0.0246) is considerably lower than the 0.048 reported. [21]. The waveguide's potential use in developing nano-photonics components such as power splitters, directional couplers, and optical bio-sensors highlights its practical significance.

II. METHODOLOGY

This research employs a simulation-based methodology using COMSOL Multiphysics, leveraging the Finite Element Method for waveguide design and performance analysis. Effective mode indices, propagation lengths, and EM field distributions are calculated. Data is analyzed via MATLAB and Origin Pro, enabling a comparative study with conventional waveguides for validation and optimization.

A. Electromagnetic Behavior of Metal According to the Drude Model

In the classical Drude model, metal is assumed to contain a large number of free electrons. Once a background of positive ion core is set against these electrons, they resemble the gas of electrons. The movement of these electrons, however, is decelerated by a peripheral electric field, causing the cores hit by electrons. The Drude model neglects all interactions during collisions. The behavior of these electrons can be modeled with Lorentz harmonic oscillator model. The

Drude Model [14], [15] as noted in Eq. (1) below, used to estimate the permittivity of metal.

$$\varepsilon = \varepsilon_{\infty} - \frac{\omega_p^2}{\omega^2 + j\omega\delta} \quad (1)$$

B. The Finite Element Method (FEM)

The Finite Element Method (FEM) is a numerical technique for solving engineering problems by discretizing structures into smaller elements. In COMSOL Multiphysics, FEM is applied to waveguide design, using material properties, boundary conditions, and discretization to solve partial differential equations. The effective mode area A_{eff} is calculated using:

$$A_{\text{eff}} = \frac{1}{\max\{W(r)\}} \int W(r) dA \quad (2)$$

where $W(r)$ is the energy density, and $A_0 = \frac{\lambda^2}{4}$ is the diffraction area in the free space. The propagation length L_p is derived from:

$$L_p = \frac{1}{2\text{Im}(\beta)} \quad (3)$$

where $\beta = n_{\text{eff}}k_0$, $k_0 = \frac{2\pi}{\lambda}$ and n_{eff} is the effective mode index. The figure of merit (FOM) is defined as:

$$\text{FOM} = \frac{L_p}{2 \sqrt{\frac{A_{\text{eff}}}{\pi}}} \quad (4)$$

Waveguide performance is optimized by varying parameters such as width, angle, and refractive indices, with results analyzed using MATLAB or Origin Pro.

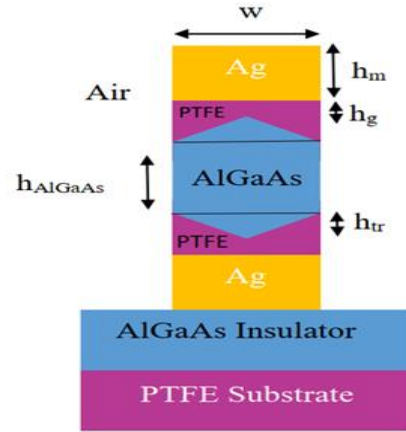


Fig. 1. Cross-sectional view of AlGaAs based triangular shape hybrid metal-insulator-metal plasmonic waveguide.

C. Model Description

The cross-sectional view of the suggested Aluminum Gallium Arsenide based TS-HMIM plasmonic waveguide has been shown in Fig. 1, where AlGaAs is used as a high index region, PTFE is used as a low index region and Silver is used as a metal. High index and Metal region are partitioned by the low index region. Relative Permittivity of AlGaAs is $\varepsilon_{\text{AlGaAs}} = 12$ and PTFE is $\varepsilon_{\text{PTFE}} = 1.7$ [14], [16]. Triangular structures in high-index (AlGaAs) regions largely contribute to increased light confinement with longer

propagation length and relatively small effective mode area. AlGaAs is a material with significant nonlinear coefficients and minimal nonlinear absorption at 1550 nm. Additionally, it is also useful for creating and utilizing nonlinear optical devices. From the Drude Model [14] relative permittivity of metal can be calculated and can be defined as:

$$\varepsilon = \varepsilon_{\infty} - \frac{\omega_p^2}{\omega^2 + j\omega\delta} \quad (5)$$

where, $\varepsilon_{\infty} = (3.7)$, $\omega_p = (1.3946 \times 10^{16} \text{ rad/s})$, and $\delta = (2.7438 \times 10^{13} \text{ s}^{-1})$ are dielectric constant at infinite angular frequency, the bulk plasma frequency, and the damping frequency for silver working at the wavelength of 1550 nm, respectively. Hence, the relative permittivity of metal (Ag) can be obtained as,

$\varepsilon_r = (-129 + 3.3i)$ at 1550 nm. Angle (γ) is considered as a result of the utilization of the triangular structure. Height of the metal, high index region (AlGaAs), low index region (PTFE), the gap between metal and high index region is denoted as respectively h_m , h_{AlGaAs} , h_{PTFE} , h_g . Here h_g denotes the gap between tip point of the high index region and metal region. Also, w is the waveguide width and h_{tr} is the height of the triangle.

D. Model Analysis and Parameter Studies

The 2-D COMSOL Multiphysics FEM-based simulator has been used in this study to analyze the parameters of the basic hybrid mode of the waveguide. This 2-D based simulator is widely used in waveguide analysis. Fig. 2. and Fig. 3. depicts the electric field distribution and its corresponding field profile, along with y- axis direction, considering the wave guide dimension of ($h_{tr} + h_{\text{AlGaAs}} + h_{tr}$) = 400 nm, $h_m = 100 \text{ nm}$, $h_g = 20 \text{ nm}$, $\gamma = 450$ and $w = 200 \text{ nm}$. In this Fig. 2, light is mainly confined at the tip point of the triangle. Fig. 3. shows the corresponding field profile in y-axis direction.

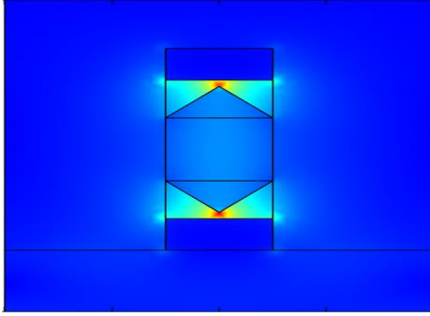


Fig. 2. Normalized Electric Field(V/m)

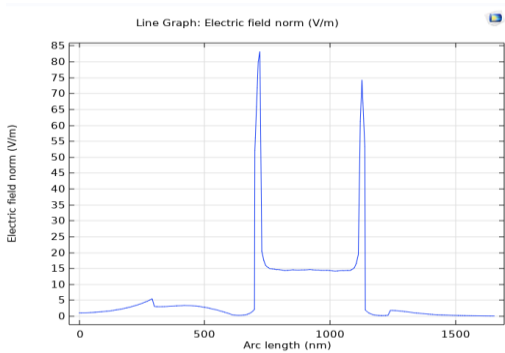


Fig. 3. Corresponding Field Profile

Using the finite-element method and the COMSOL software, we first present a detailed geometrical diffraction-limited area in free space and equals $\lambda^2/4$. Despite the fact that plasmonic structures can produce electromagnetic fields that are incredibly localized, the metal's internal damping limits how far the field can travel. The propagation distance L_p which is defined as the distance at which the amplitude parameter study of the proposed waveguide structure based on two key indicators: normalized effective mode area (A_{eff}/A_0) and propagation length (L_p). The modal area measurements that are currently available come from traditional waveguide theory. In order to consistently quantify the mode confinement, we need to find a definition that takes into account the full extent of the plasmonic field distribution. By using the Eq. (2) we can calculate the normalized effective mode area. The normalized effective mode area is A_{eff}/A_0 where A_0 is the of the energy density attenuates to $1/e$. By the Eq. (3) we calculated the propagation length. Figure of merit (FOM) can be calculated from the Eq. (4) which show the quick performance of the proposed waveguide structure.

III. RESULT

A. Effect of Height of Metal on Fundamental Mode Profile

In this subsection height of the metal is varied to investigate the performance of the proposed plasmonic waveguide in fundamental mode profile at the waveguide dimension of $w = 200 \text{ nm}$, of $h_{tr} + h_{\text{AlGaAs}} + h_{tr} = 400 \text{ nm}$, $h_g = 20 \text{ nm}$, $\gamma = 450$. This value of the waveguide dimension is fixed, only varying parameter is height of the metal (h_m). Metal height is varied from 50 nm to 400 nm for each $h_g = 5 \text{ nm}$, 10 nm and 20 nm . Fig. 4. depicts that Real Part of Effective Refractive Index ERI_{real} initially decreasing with the increase of metal height h_m and $h_m > 200$ its gets saturated. Also depicts that for smaller h_g , ERI_{real} value is larger on account of the effect of traditional plasmonic nature of waveguide (Ag-AlGaAs). The propagation length of the current waveguide, as shown in Fig. 5, increases with the rise in metal height until reaching 200 nm, after which it stabilizes. Similarly, in Fig. 7. the figure of merit exhibits the same trend, increasing with metal height.

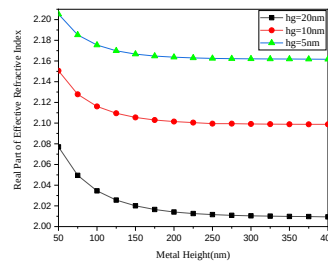


Fig. 4. Variation in effective refractive index with Metal Height (hm)

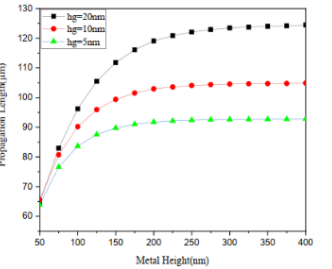


Fig. 5. Variation in propagation length with Metal Height (hm)

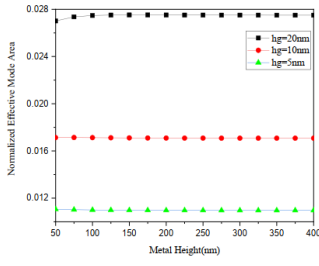


Fig. 6. Variation in normalized effective mode area with Metal Height (hm)

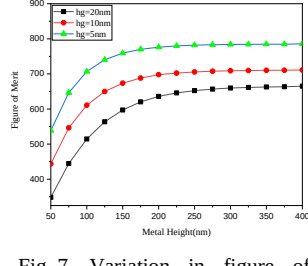


Fig. 7. Variation in figure of merit with Metal Height (h_m)

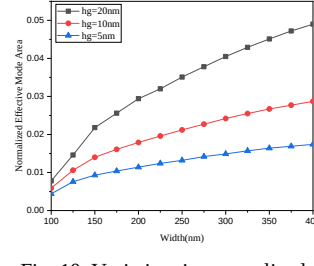


Fig. 10. Variation in normalized effective mode area with Width(w)

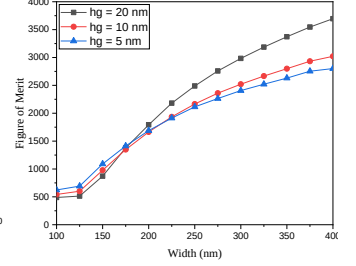


Fig. 11. Variation in figure of merit with Width(w)

B. Effect of Waveguide Width on Fundamental Mode Profile

In this segment, it has been determined, for all the considered gaps (i.e $h_g=20\text{nm}$, 10nm , 5nm) taken into consideration and show the effects of waveguide width on the fundamental mode profile in the suggested hybrid plasmonic waveguide. Here, the values of h_m , $h_{tr} + h_{AlGaAs} + h_{tr}$, γ has been fixed respectively as, 100 nm , 550 nm , and 450nm . In this subsection only varying parameter is waveguide width. Width is varied from 100nm to 400 nm for each considered gap. From the Fig. 8. it is shown that real part increases with the increase of waveguide width up to 400 nm . From the Fig. 9, It is evident that the propagation length of the current waveguide grows as the waveguide width expands from 125 nm up to 400 nm . From the figure it has been observed that for larger h_g more propagation length can be achieved. This is due to the fact that the HPW is more affected by conventional plasmonic structure as a result of Ag-AlGaAs. Additionally, Fig. 10. illustrates the direct proportionality between the waveguide width (w) and A_{eff}/A_0 . More energy may be contained in the low-index zone as w increases, which results in low loss and longer propagation length. Fig. 11. illustrated the variation in figure of merit with Width(w) from 100 nm to 400 nm similar trends like 4.6 where initially figure of merit with the waveguide width don't change too much.

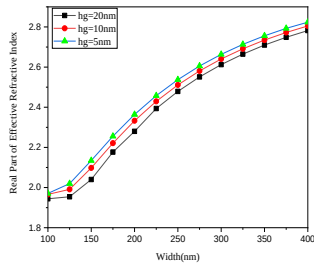


Fig. 8. Variation in effective refractive index with Width(w)

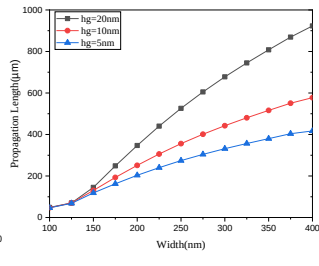


Fig. 9. Variation in propagation length with Width(w)

C. Effect of Angle of Triangular Shape on Fundamental Mode Profile

In this portion the angel of the triangular shape (γ) is varied from 5° to 50° to investigate the performance of the proposed plasmonic waveguide in fundamental mode profile at the waveguide dimension of $w=200\text{ nm}$, $h_{tr} + h_{AlGaAs} + h_{tr} = 400\text{nm}$, $h_m=100\text{nm}$ for all considered gaps ($h_g=20\text{nm}, 10\text{nm}, 5\text{nm}$) between high index tip point to metal layer. Fig. 12. depicts the variation in ER_{real} with the angle (γ). From this figure it is clear that with the increasing in (γ) Real Part of Effective Refractive Index decreases. Fig. 13. shows the relationship between Propagation length with angle (γ). This Fig. 13. depicts that initially propagation length increases with the angel (γ) but when $\gamma > 30$, it shows the decreasing tendency. Also, this figure depicts that for higher h_g propagation length is maximum. Fig. 4.14. Normalize Effective Mode area and Fig. 4.12 are opposite trends with angle. For all h_g values, the FOM initially increases with the angle (γ), reaches a peak, and then decreases as the angle increases further.

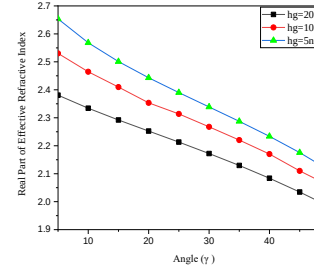


Fig. 12. Variation in effective refractive index with Angle (γ)

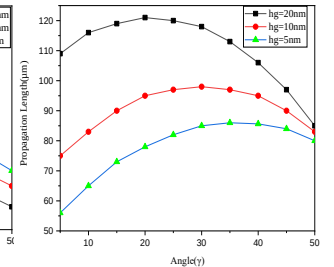


Fig. 13. Variation in propagation length with Angle(γ)

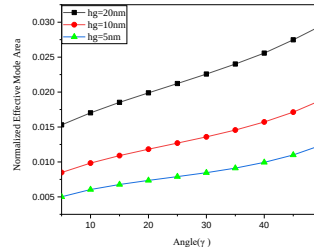


Fig. 14. Variation in normalized effective mode area with Angle(γ)

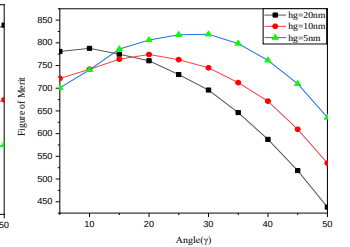


Fig. 15. Variation in figure of merit with Angle(γ)

D. Effect of Height of High-Index (AlGaAs) Region on Fundamental Mode Profile

This portion examines the impact of varying the height of the high-index region on the fundamental mode profile in the TS-HMIM plasmonic waveguide. The study explores

changes in h_{AlGaAs} , increasing it from 100 nm to 400 nm, across all considered gap sizes ($h_g=20$ nm, 10 nm, 5 nm). For the angle 45° , h_m , w , and h_{tr} have been set to 100 nm, 200 nm, and 100 nm, respectively. According to the analysis Fig. 16. shows the relation between ERI_{real} to the height of high index region (i.e. h_{AlGaAs}). From this figure it is noticeable that with the increase in h_{AlGaAs} the value of ERI_{real} increases. Fig. 17. shows the variation in propagation length with high index region. This figure depicts that propagation length increases exponentially with the height of high index region. From the figure it is clear that larger h_g provides larger propagation length. From the Fig. 18. it is clear that effective mode area increases with the increase in height of high index AlGaAs zone and more energy can be confined. Fig. 19. experienced almost similar trends with height of AlGaAs, where FOM increases exponentially.

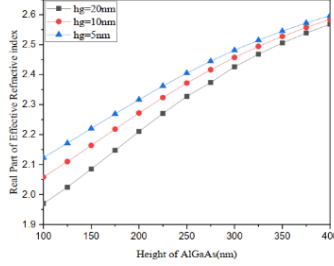


Fig. 16. Variation in effective refractive index with Height of AlGaAs (h_{AlGaAs})

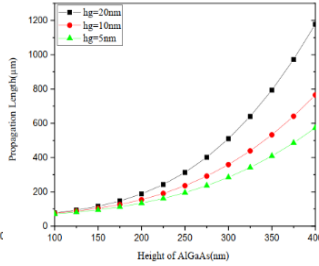


Fig. 17. Variation in propagation length with Height of AlGaAs (h_{AlGaAs})

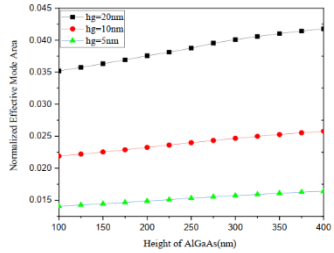


Fig. 18. Variation in normalized effective mode area with Height of AlGaAs (h_{AlGaAs})

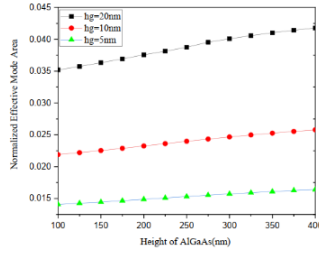


Fig. 19. Variation in figure of merit with Height of AlGaAs (h_{AlGaAs})

E. Comparison of Current Work with Recent Literature

TABLE I. COMPARISON OF THE WORK

References	ERI_{real}	Propagation Length (μm)	A_{eff}/A_0	FOM
[17]	--	40	0.045	--
[18]	--	45	0.13	150
[19]	2.38	77.1	0.076	--
[1]	--	95	0.125	--
[20]	2.09	145.3	0.054	--
[14]	--	188.2	0.059	--
[21]	--	228	0.125	--
[22]	2.32	279.7	0.048	1452
[23]	--	319.00	0.0558	--
Proposed Work	2.45	359.01	0.0246	2025

The proposed work demonstrated in Table. I significant superiority over the prior literature in several key performance metrics. With an ERI_{real} value of 2.45, it exceeds all the references, such as [19] (2.38) and [22] (2.32), showcasing

enhanced optical properties. In terms of Propagation Length, the proposed work achieves an impressive 359.01 μm , which is notably higher than all the other references, such as [23] (319.00 μm) and [21] (228 μm), indicating minimal signal attenuation and superior performance in signal transmission. Additionally, the A_{eff}/A_0 ratio of the proposed work is 0.0246, which is smaller than most references, such as [19] (0.076) and [20] (0.054), implying better mode confinement and higher efficiency. Most notably, the proposed work achieves a Figure of Merit (FOM) of 2025, far surpassing the next highest value of 1452 in [22] and 150 in [18]. These results clearly highlight the superior performance of the proposed method across all key metrics, establishing its effectiveness and advancements over existing approaches. field.

IV. CONCLUSION

In this study, a novel Triangular Shape Hybrid Metal-Insulator-Metal (HIM) Plasmonic Waveguide has been designed and analyzed using COMSOL Multiphysics, focusing on reducing propagation loss and achieving high light confinement for subwavelength propagation. The proposed waveguide, utilizing Silver (Ag), Aluminum Gallium Arsenide (AlGaAs), and Polytetrafluoroethylene (PTFE) materials, has demonstrated a significant improvement in performance with a propagation length of 359.01 μm and a normalized effective mode area of 0.0246. This design outperforms previous works in terms of propagation length and mode confinement, as evident in the comparative analysis with recent literature. The results indicate that higher AlGaAs heights contribute to increased propagation lengths, making this waveguide suitable for nano-phonic applications with low attenuation and long-range signal propagation.

This research also addresses the critical issue of propagation loss in plasmonic waveguides by proposing a hybrid approach that combines the benefits of both plasmonic and dielectric waveguides. The findings suggest that this hybrid plasmonic waveguide could be an ideal candidate for integration with active optical devices, offering a promising solution for applications such as power splitters, directional couplers, and optical bio-sensing. The work presents an effective approach to optimizing the trade-off between propagation length and normalized effective mode area, offering new opportunities for advancing nano-phonic technologies.

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